



Bank Marketing Campaign: Predictive Analytics

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Business Problem

- Banks spend significant resources on marketing campaigns, but most calls end without success. This leads to wasted time, money, and reduced customer trust.
- The main problem is how to predict which clients are more likely to subscribe to a term deposit, so that campaigns can be more targeted and cost-effective.
- Stakeholders Include: Bank management, Marketing teams, Customers



Research Questions

1. What customer demographic and financial characteristics influence subscription decisions?
2. How do campaign-related variables affect customer response?
3. Can we build a model that accurately predicts whether a client will subscribe to a term deposit?
4. Which features are the most important for prediction?

Data Overview

- **Dataset:** Bank Marketing – UCI Repository (bank-additional-full.csv)
- **Records:** 45,211
- **Features:** Age, job, marital status, education, default, balance, loans, contact type, campaign details, previous outcomes
- **Target:** y = client subscription status
Yes → client subscribed
No → client did not subscribe
- **Purpose:** Predict whether a client will subscribe to a term deposit

Methodology

- Data Collection: Bank Marketing dataset (41,188 records, 20 attributes)
- Preprocessing: Handle missing values, encode categorical data, scale features, split train/test
- Modeling: Logistic Regression, Random Forest, XGBoost
- Evaluation: Accuracy, Precision, Recall, F1-Score
- Insights: Feature importance analysis & campaign optimization recommendations



Objectives

Main Objective: To build a predictive classification model that determines whether a client will subscribe to a term deposit.

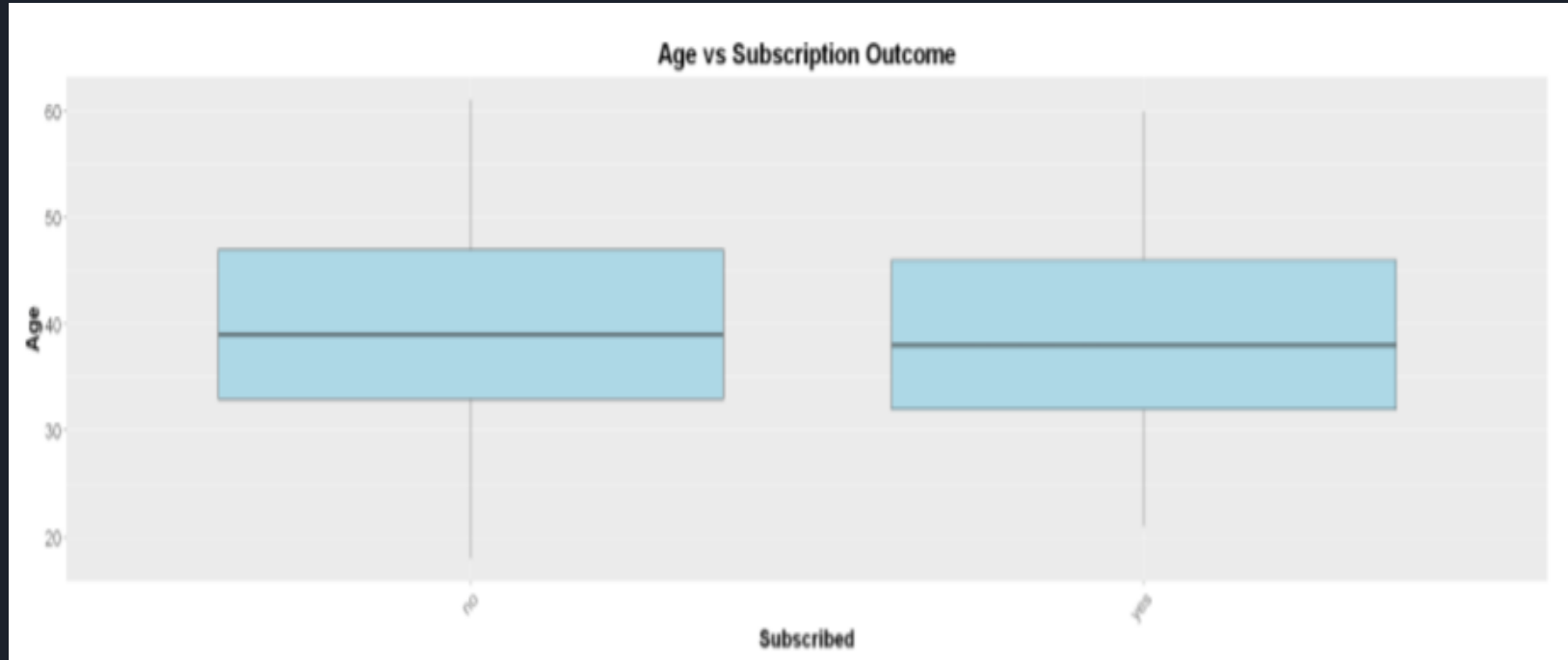
Specific Objectives:

Perform data preprocessing (cleaning, handling missing values, encoding).

Conduct exploratory data analysis (EDA) to identify trends and patterns.

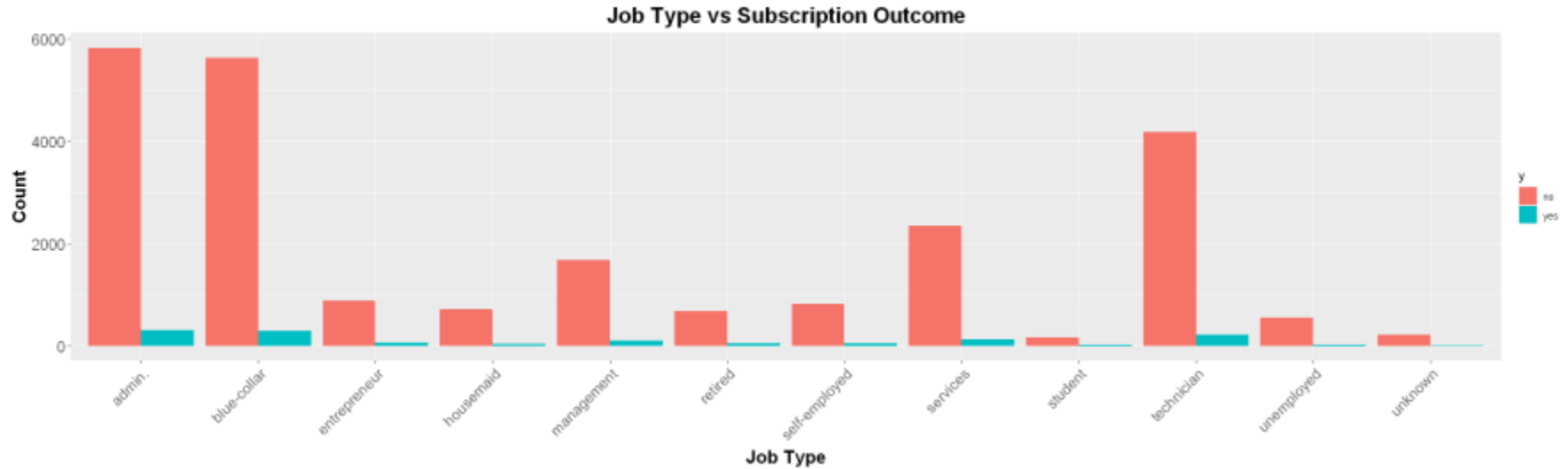
Provide conclusions and recommendations for banking decision-makers.

Research Question 1 – Customer Characteristics



Observations: Subscription rates are higher among middle-aged, married, and educated clients

Research Question 1 – Customer Characteristics



Observations:

Blue-collar and management roles dominate in volume but have low-to-high subscription conversion.

Retired and unemployed groups show the lowest subscription interest.

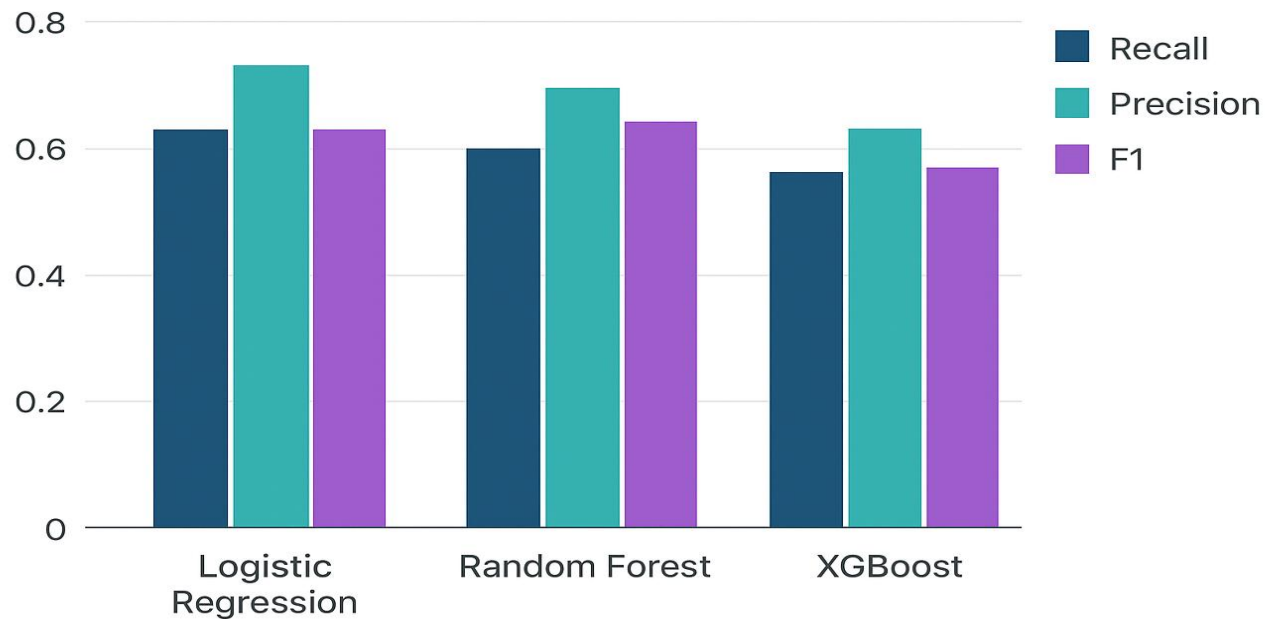
Research Question 2 – Campaign Variables



Most individuals received fewer contacts, with distributions concentrated around 1–2 contacts for both subscription outcomes. Subscribed clients show slightly more spread, suggesting that multiple contacts may improve conversion likelihood.

Model Performance and Evaluation

Model Performance Overview





Model Selection

- It gives the most balanced results overall.
- It misses fewer important cases than Logistic Regression.
- It is more reliable and consistent than XGBoost.



Conclusion

Customer Characteristics:

Age, education level, and financial status (e.g., housing or personal loan) influence subscription. More educated clients showed higher subscription likelihood.

Campaign Variables:

Contact duration and the number of recent contacts were strong predictors—longer and more recent interactions increased subscription chances.

Economic Factors:

Indicators such as employment variation rate and consumer confidence had subtle but notable effects on customer responsiveness.



Recommendations

Prioritize High-Probability Customers

Improve Contact Strategies

Focus on contacts that have longer durations or recent activity to increase engagement success.

Use Predictive Models in Campaigns

Integrate high-performing models like Random Forest into marketing workflows to optimize targeting and maximize ROI.

Optimize Timing Using Economic Signals